Australian National University
Research School of Earth Sciences

Groundwater Overview

EMSC3025/6025 — Recap Session

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Objectives of this Recap Session

By the end of this session, you should be able to:

- **Define** key groundwater parameters: porosity, hydraulic conductivity, transmissivity, and storativity
- **Apply** Darcy's Law to calculate flow rates and velocities in different media
- **Distinguish** between confined, unconfined, and perched aquifer systems
- **Analyze** characteristic curves for unsaturated flow ($\theta(\psi)$, $K(\psi)$, $C(\psi)$)
- **Solve** practical groundwater problems using flow theory and boundary conditions
- **Evaluate** real-world aquifer systems and their management challenges

The Big Picture: Groundwater in the Global Context

- Groundwater represents **1.69%** of Earth's total water
- **23,400 × 10^3 km^3** stored globally
- **Most accessible freshwater** for human use
- Critical for:
  - **Drinking water supply** (billions depend on it)
  - **Agricultural irrigation** (40% of global food production)
  - **Industrial processes**
  - **Ecosystem support** (baseflow to rivers)

Challenge: Invisible, slow-moving, easily overexploited

Region	Groundwater Use (%)
Australia	21% of total water
USA	26% of total water
India	60% of irrigation
Europe	75% of drinking water

Table 1: Groundwater dependence varies globally

Fundamental Parameters: The Building Blocks

Porosity: The Foundation of Groundwater Storage

Total porosity describes void space available for water storage:

$$n_T = \frac{V_v}{V_T} = \frac{V_T - V_s}{V_T}$$

Alternative calculation using bulk density: $n_T = 1 - \frac{\rho_b}{\rho_s}$

Two types:

- **Primary porosity**: Original voids (sediment packing, vesicles)
- **Secondary porosity**: Later formation (fractures, dissolution)

Typical porosity values:

Material	Porosity Range
Sand	0.25 – 0.35
Silt	0.35 – 0.50
Clay	0.40 – 0.55
Sandstone	0.05 – 0.25
Limestone	0.01 – 0.30
Granite	<0.01

Table 2: Porosity varies dramatically by material

Effective Porosity vs Total Porosity

Not all pores contribute to flow:

- **Total porosity** (n_T): All void space
- **Effective porosity** (n_e): Connected pores that allow flow
- **Relationship:** $n_e \leq n_T$

In fractured media:

- **Matrix porosity**: Storage in rock blocks
- **Fracture porosity**: Flow pathways
- **Dual porosity systems**: Both contribute

Key insight

High porosity ≠ High permeability

Clay: High porosity, low permeability
Fractured granite: Low porosity, high permeability

Hydraulic Conductivity: The Flow Parameter

Hydraulic conductivity (K) quantifies ease of water transmission:

$q = -K\frac{dh}{dl}$ (Darcy's Law)

Units: Length/time (m/day, cm/s)

Depends on:

- **Medium properties**: Pore size, connectivity, tortuosity
- **Fluid properties**: Density, viscosity

Intrinsic permeability (k): Medium-only property $K = \frac{k\rho_w g}{\mu}$

Hydraulic conductivity ranges:

Material	K (m/s)
Gravel	10^{-1} – 10^{-3}
Sand	10^{-3} – 10^{-6}
Silt	10^{-6} – 10^{-9}
Clay	10^{-9} – 10^{-12}
Fractured rock	10^{-5} – 10^{-8}
Intact rock	10^{-10} – 10^{-14}

Table 3: 12+ orders of magnitude variation!

Measuring Hydraulic Conductivity

Laboratory methods:

Constant-head test (coarse materials): $K = \frac{QL}{Ah}$

- Steady flow through sample
- Good for sands, gravels

Falling-head test (fine materials): $K = 2.3 \frac{aL}{A(t_1 - t_0)} \log_{10} \left(\frac{h_0}{h_1} \right)$

- Declining head over time
- Good for clays, silts

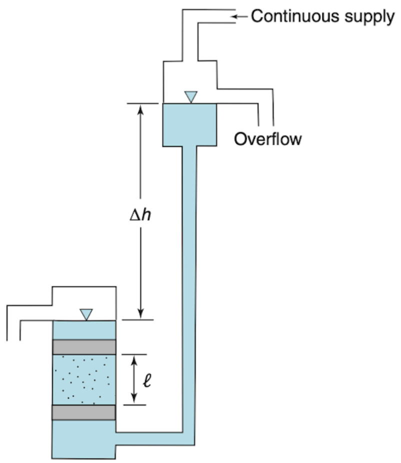


Figure 1: Constant-head permeameter setup

Field methods:

- Slug tests
- Pumping tests
- Tracer tests

Hydraulic Head: The Driving Force

Hydraulic head represents energy available to move groundwater:

$$h = z + \frac{P}{\rho_w g}$$

Components:

- **Elevation head** (z): Height above datum
- **Pressure head** ($P/\rho_w g$): Pressure energy
- **Velocity head**: Usually negligible in groundwater

Field measurement: $h = EL_{TOC} - DTW$

- EL_{TOC}: Elevation of top of casing
- DTW: Depth to water

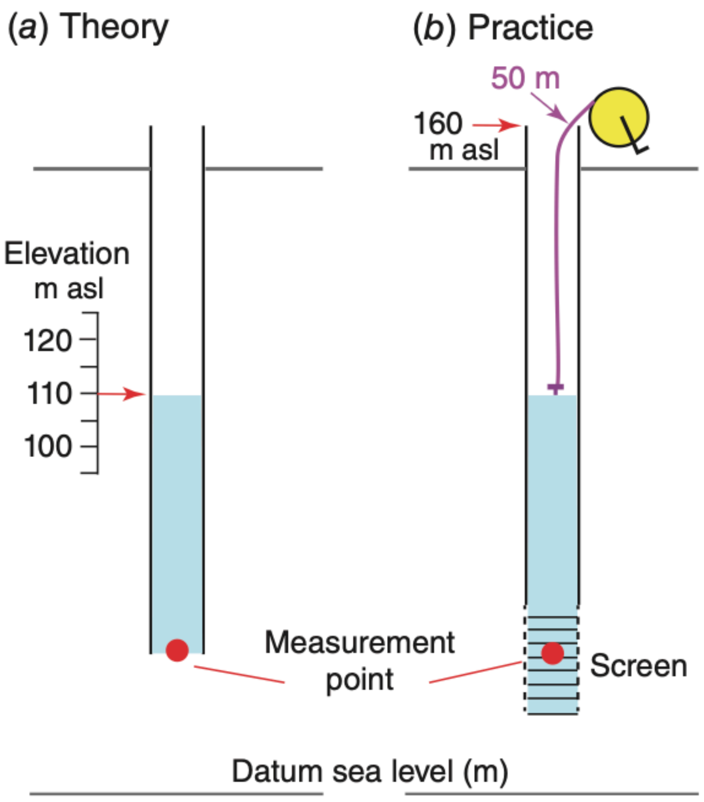


Figure 2: Measuring hydraulic head with piezometers

Flow direction: From high head to low head

Gradient: $i = \frac{dh}{dl}$

Aquifer Properties: Storage and Transmission

Transmissivity: Aquifer-Scale Flow Parameter

Transmissivity integrates conductivity over aquifer thickness:

$$T = K \times b$$

Where:

- T = transmissivity [L^2/T]
- K = hydraulic conductivity [L/T]
- b = aquifer thickness [L]

Darcy's Law for aquifers: $Q = T \times W \times i$

- Q = discharge [L^3/T]
- W = width perpendicular to flow
- i = hydraulic gradient

Physical meaning

Transmissivity tells you how much water can flow through the **entire thickness** of an aquifer per unit width.

Units: m^2/day, ft^2/day

Storativity: Water Release from Storage

Storativity (S) quantifies water release per unit area per unit head change:

$$S = \frac{\text{volume of water}}{\text{unit area} \times \text{unit head change}}$$

Dimensionless quantity

Relationship to specific storage: $S = S_s \times b$

- S_s = specific storage [1/L]
- b = aquifer thickness [L]

Typical storativity values:

Aquifer Type	S Range
Confined	10^{-5} – 10^{-3}
Unconfined	0.01 – 0.30

Table 4: 3+ orders of magnitude difference!

Why the difference? Different storage mechanisms!

Storage Mechanisms: Confined vs Unconfined

Confined aquifer storage:

Two mechanisms supply water:

1. **Water expansion**: Lower pressure → molecules expand
2. **Aquifer compression**: Reduced support → grains compress

$$S_s = \rho_w g(\beta_p + n\beta_w)$$

- β_p = matrix compressibility
- β_w = water compressibility
- Matrix compression dominates

Unconfined aquifer storage:

Early time: Elastic mechanisms (like confined) **Later time**: **Gravity drainage** dominates

$$S = S_y + bS_s$$

Specific yield (S_y) >> bS_s

- S_y = drainable porosity
- Much more water released per head drop

Specific Yield and Specific Retention

Specific yield (S_y): Water released by gravity drainage

$$S_y = \frac{V_d}{V_T}$$

Specific retention (S_r): Water retained against gravity

$$S_r = \frac{V_r}{V_T}$$

Fundamental relationship: $n = S_y + S_r$

Material characteristics:

Material	S_y	S_r
Coarse gravel	0.20-0.25	0.01-0.05
Fine sand	0.10-0.25	0.05-0.15
Silt	0.05-0.15	0.15-0.30
Clay	0.01-0.05	0.30-0.50

Table 5: Inverse relationship: High $ S_r $ → Low $ S_

Aquifer Types and Classification

Aquifer Classification by Confinement

Three main types:

1. Unconfined (water table) aquifers:

- Water table forms upper boundary
- No confining layer above
- Water under atmospheric pressure at water table

2. Confined (artesian) aquifers:

- Bounded by confining beds above and below
- Water under pressure > atmospheric
- Water level in wells above aquifer top

3. Perched aquifers:

- Unconfined aquifer above regional water table
- Forms on low-permeability layer
- Unsaturated zone exists below

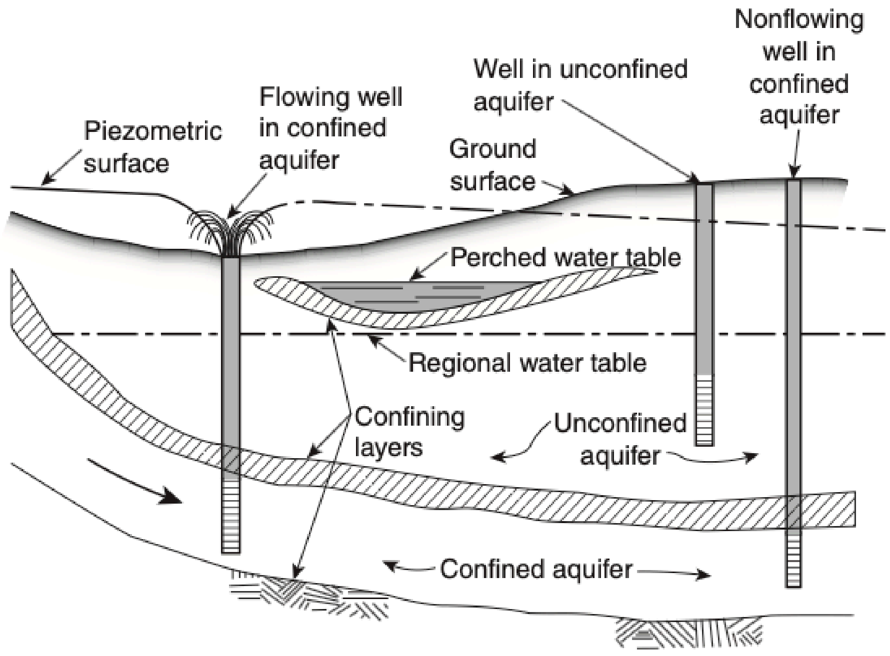


Figure 3: Unconfined, confined, and perched aquifers

Confining Beds: More Than Just Barriers

Traditional classification:

- **Aquifuge**: No storage, no transmission (intact crystalline rock)
- **Aquiclude**: Storage but no transmission (clay layers)
- **Aquitard**: Some storage and transmission (silty clay)

Modern understanding: Aquitardifers

- **Extreme anisotropy**: $K_h >> K_v$
- **Horizontal flow**: Like an aquifer
- **Vertical resistance**: Like a confining bed
- **Fractures**: Can create fast pathways

Real-world complexity

Simple textbook models often don't capture the dual nature of confining beds.

Management implications:

- Detailed site characterization essential
- Consider both protective and transmissive functions

Unsaturated Zone: Where Constants Become Functions

The Unsaturated Zone Challenge

Key insight: In unsaturated media, hydraulic **constants** become hydraulic **functions**

Two-phase flow: Air + water in pores

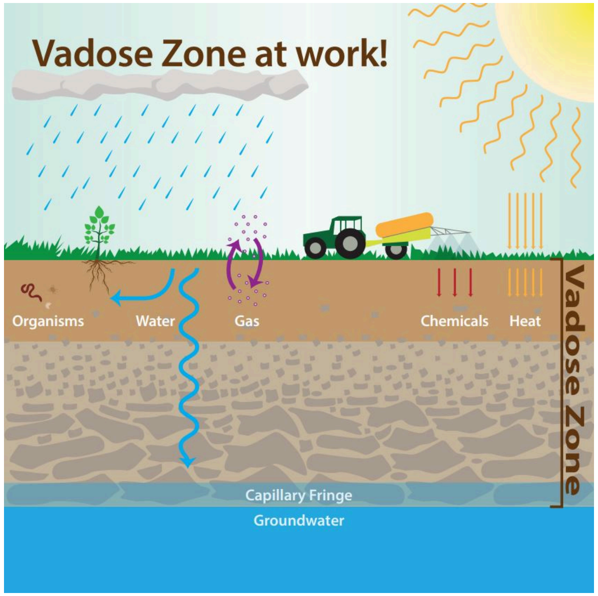
- As soil dries → void spaces fill with air
- Harder for water to flow
- Both θ and K depend on ψ

Pressure head in unsaturated zone:

- $\psi < 0$ (suction/tension)
- Water pressure less than atmospheric
- Explains why water doesn't flow to boreholes

Fundamental change

Saturated: K = constant Unsaturated: $K(\psi)$ = function



Three Characteristic Curves

Three fundamental relationships describe unsaturated flow:

1. **Water Retention Curve** $\theta(\psi)$: How soil holds/releases water as pressure changes
2. **Hydraulic Conductivity Curve** $K(\psi)$: How flow capacity varies with water content
3. **Moisture Capacity Curve** $C(\psi)$: How storage properties change with pressure

All relationships are strongly nonlinear and depend on soil type and pore-size distribution.

Water Retention Curves θ(ψ)

Water retention curve: Relationship between negative pressure head and volumetric water content

- As θ decreases → ψ becomes more negative
- Curves are **nonlinear** regardless of scaling
- At extremes: small θ change → large ψ change
- **Hysteresis**: Different curves for drying vs wetting

Effective saturation normalizes for soil type:

$$\boxed{s_e = \frac{\theta - \theta_r}{n - \theta_r}}$$

Two main models:

Brooks-Corey: $s_e = \left(\frac{\psi_b}{\psi}\right)^{\lambda}$ (if $\psi < \psi_b$)

van Genuchten: $s_e = \frac{1}{[1+(\alpha|\psi|)^{\beta}]^{\gamma}}$

Both use effective saturation s_e (0 to 1)

Unsaturated Hydraulic Conductivity K(ψ)

Relative hydraulic conductivity shows fraction of saturated value:

$$K_r(\psi) = \frac{K(\psi)}{K_s}$$

Brooks-Corey model: $K_r(\psi) = \left(\frac{\psi_b}{\psi}\right)^{2+3\lambda}$ (if $\psi < \psi_b$)

van Genuchten model: $K_r(\psi) = \frac{[1-(\alpha|\psi|)^{\beta-1}(1+(\alpha|\psi|)^{\beta})^{-\gamma}]^2}{[1+(\alpha|\psi|)^{\beta}]^{\gamma/2}}$

Key behavior

Near saturation: $K_r \approx 1$ As soil dries: $K_r \to 0$

Strongly nonlinear decrease

Moisture Capacity C(ψ)

Moisture capacity: Slope of water retention curve

$$c_m = \frac{d\theta}{d\psi}$$

Physical meaning: How much θ changes for given ψ change

Storage in unsaturated zone: Combines moisture capacity and specific storage $(c_m + sS_s)\frac{\partial h}{\partial t}$

Applications:

- **Richards' equation** for unsaturated flow
- **Infiltration modeling**
- **Vadose zone transport**

Derivative relationship

c_m derived from $\theta(\psi)$ models using calculus

Groundwater Flow Theory and Applications

Darcy's Law: The Foundation

Darcy's Law relates flow to energy gradients:

$$q = -K\frac{dh}{dl}$$

Physical meaning:

- Water flows from high to low energy (head)
- Flow rate proportional to gradient and conductivity
- Negative sign: flow opposes gradient direction

3D form: $q_x = -K_x\frac{\partial h}{\partial x}, q_y = -K_y\frac{\partial h}{\partial y}, q_z = -K_z\frac{\partial h}{\partial z}$

Linear (pore) velocity:

$$v = \frac{q}{n_e}$$

Key insight: Actual water speed > Darcy velocity because water flows only through connected pores

The Groundwater Flow Equation

Fundamental equation for saturated flow (from mass conservation + Darcy's Law):

$$\frac{\partial}{\partial x}\left(K_x \frac{\partial h}{\partial x}\right) + \frac{\partial}{\partial y}\left(K_y \frac{\partial h}{\partial y}\right) + \frac{\partial}{\partial z}\left(K_z \frac{\partial h}{\partial z}\right) = S_s \frac{\partial h}{\partial t}$$

Special cases:

- **Steady-state:** $\frac{\partial h}{\partial t} = 0$
- **Isotropic, homogeneous:** Laplace equation $\nabla^2 h = 0$

What it describes:

- How hydraulic head varies in space and time
- 3D flow in heterogeneous, anisotropic media
- Foundation for all groundwater modeling

Boundary Conditions: Defining the Problem

Three types of boundary conditions:

1. Dirichlet (specified head): $h = h_1(x, y, z, t)$ at boundary

- Large water bodies (lakes, rivers)
- Known, stable water levels

2. Neumann (specified flux): $q_n = q_n(x, y, z, t)$ at boundary

- Recharge zones ($q_n > 0$)
- Impermeable boundaries ($q_n = 0$)
- Pumping wells ($q_n < 0$)

3. Cauchy (head-dependent flux): $q_n = \frac{K_m}{M}[h_m - h]$

- River-aquifer interactions
- Semi-permeable boundaries
- Flow direction can reverse

Critical point

Boundary conditions determine the unique solution. Same equation, different boundaries = different answers.

Flow Nets: Visualizing Groundwater Flow

Flow nets provide graphical solutions to Laplace equation:

Two key components:

- **Streamlines**: Show flow direction
- **Equipotential lines**: Contours of equal head

Key properties:

- Streamlines ⊥ equipotentials (in isotropic media)
- Form curvilinear squares
- Equal head drop between equipotentials
- Same flow between streamlines

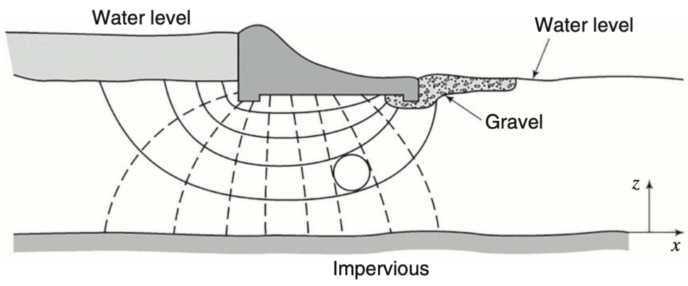


Figure 4: Dam seepage flow net

Applications:

- Dam seepage analysis
- Well capture zones
- Contaminant transport paths

Real-World Aquifer Systems

Major Aquifer Types and Characteristics

Aquifer Type	Porosity	K Range	Storage	Key Feature	Main Challenge
Unconsolidated	High (0.25-0.35)	High	$S_y = 0.1 - 0.3$	Intergranular	Contamination
Basin-fill	High	High-Moderate	High	Thick accumulations	Over-exploitation
Sandstone	Low-Moderate	Low-Moderate	$S = 10^{-3} - 10^{-5}$	Fracture flow	Vertical leakage
Carbonate	Variable	10 orders of magnitude	Variable	Solution conduits	Heterogeneity
Karst	Highly variable	Extremely high locally	Low-Moderate	Caves and conduits	Rapid transport
Crystalline	Very low	Very low	Very low	Fractures only	Limited yield

Table 6

Unconsolidated Aquifers: High Yield Systems

Characteristics:

- **Intergranular porosity**: 0.25-0.35
- **High permeability**: Varies with sorting
- **Unconfined conditions**: Direct recharge
- **High specific yield**: 0.1-0.3
- **Most extensively pumped** in many regions

Four main categories:

1. **Basin-fill**: Central Valley (CA), Arizona basins
2. **Blanket sand/gravel**: High Plains aquifer
3. **Glacial deposits**: Northern US, Canada
4. **Stream-valley**: Local, small extent

Advantages:

- High storage and transmission
- Predictable flow patterns
- Good well yields

Challenges:

- Susceptible to contamination
- Over-exploitation risks
- Land subsidence potential

Consolidated Rock Aquifers: Complexity and Heterogeneity

Sandstone aquifers:

- **Cemented sand**: Reduced porosity
- **Fracture/bedding flow**: Main pathways
- **Large areal extent**: Regional water supply
- **Example**: Dakota Sandstone (46% recharge through confining beds)

Carbonate aquifers:

- **Solution enhancement**: Dissolution creates conduits
- **Extreme heterogeneity**: 10 orders of magnitude K range
- **Large yields**: Where well-connected
- **Example**: Floridan aquifer system

Crystalline rock aquifers:

- **Fractured granite/basalt**: Very low matrix porosity
- **Flow in fractures only**: Secondary porosity controls
- **Highly variable yields**: Depends on fracture networks
- **Important**: New England, Canadian Shield

Volcanic rock aquifers:

- **Basalt flows**: Cooling fractures and lava tubes
- **Vesicular zones**: Water storage
- **Examples**: Hawaiian volcanic, Columbia Plateau

Karst Systems: Extreme Heterogeneity

Formation requirements:

1. **Chemically aggressive groundwater** (low pH)
2. **Fractures present** for transmission
3. **Groundwater can drain out** (hydraulic gradient)

Solution features evolution:

- Start as small conduits (~1 cm)
- Grow to cave size (kilometers long)
- **Highest yields** at fracture intersections
- **Extremely heterogeneous** permeability

Karst landforms:

- **Sinkholes**: Dissolution, subsidence, collapse
- **Stream sinks**: Disappearing streams
- **Dry valleys**: Former surface drainage
- **Large springs**: Concentrated discharge
- **Cave networks**: Underground drainage

Management challenges:

- Rapid contaminant transport
- Unpredictable flow paths
- Surface-subsurface connectivity

Practical Applications: Tutorial Insights

Problem-Solving Approaches: Key Calculations

Hydraulic head and flow direction:

- Calculate hydraulic gradient: $i = \frac{h_1 - h_2}{\Delta l}$
- Determine flow direction: High head → Low head
- Apply Darcy's Law: $q = -K \cdot i$

Linear groundwater velocity:

- Darcy velocity: $q = K \cdot i$
- Pore velocity: $v = \frac{q}{n_e}$
- **Key insight**: $v > q$ because flow through connected pores only

Pressure and head calculations:

- Total head: $h = EL_{TOC} - DTW$
- Pressure head: $\psi = h - z$
- Absolute pressure: $P = \rho_w g(h - z)$

Intrinsic permeability:

- Convert from hydraulic conductivity
- $k = \dfrac{K\mu}{\rho_w g}$
- Useful for fluid-independent comparisons

Layered Aquifer Systems: Equivalent Properties

Horizontal equivalent conductivity (thickness-weighted average):

$$K_h = \frac{\sum_{i=1}^{M} b_i K_{hi}}{\sum_{i=1}^{M} b_i}$$

Vertical equivalent conductivity (harmonic average):

$$K_v = \frac{\sum_{i=1}^{M} b_i}{\sum_{i=1}^{M} \frac{b_i}{K_{vi}}}$$

Flow refraction across boundaries: $\frac{K_1}{K_2} = \frac{\tan(\alpha_1)}{\tan(\alpha_2)}$

Key insight

Low-K layers dominate vertical resistance (harmonic average)

High-K layers dominate horizontal flow (arithmetic average)

Boundary Condition Applications

Identifying boundary types:

- **Constant head**: Large water bodies, known levels
- **No-flow**: Impermeable boundaries, symmetry lines
- **Specified flux**: Recharge zones, pumping wells
- **Head-dependent**: River-aquifer interactions

Flow net construction:

1. Identify boundary conditions
2. Sketch streamlines and equipotentials
3. Ensure orthogonality and square formation
4. Count flow channels and head drops

Analytical solutions:

- **Confined aquifer**: Linear head distribution $h = h_0 + (h_L - h_0)\frac{x}{L}$
- **Unconfined aquifer**: Curved water table $h = \sqrt{h_0^2 + (h_L^2 - h_0^2)\frac{x}{L}}$

Flow rate calculations:

- Use appropriate form of Darcy's Law
- Account for aquifer geometry
- Consider boundary conditions

Case Studies: Real-World Applications

Dam seepage analysis:

- Flow nets reveal seepage patterns
- Identify high-gradient zones
- Assess uplift pressures
- Design drainage systems

Aquifer characterization:

- Pumping test analysis
- Tracer test interpretation
- Multi-well monitoring networks
- Potentiometric surface mapping

Contamination assessment:

- Determine flow directions
- Estimate travel times
- Design monitoring networks
- Evaluate remediation options

Water supply management:

- Sustainable yield calculations
- Well interference analysis
- Conjunctive use planning
- Climate change impacts

Integration: Putting It All Together

The groundwater system approach:

- **Physical properties**: Porosity, permeability, storage
- **Flow dynamics**: Darcy's Law, gradients, boundaries
- **System behavior**: Confined vs unconfined responses
- **Unsaturated zone**: Variable properties, characteristic curves
- **Real-world complexity**: Heterogeneity, anisotropy, scale effects

Problem-solving strategy:

1. **Define the system**: Geometry, boundaries, properties
2. **Identify governing equations**: Steady vs transient, dimensions
3. **Apply boundary conditions**: Match physical constraints
4. **Solve or approximate**: Analytical, numerical, or graphical
5. **Interpret results**: Physical meaning, limitations, implications

Satellite Geodesy: Remote Sensing of Water

Remote Sensing of Water: Key Concepts for Exam

Critical distinction for exam success:

What satellites MEASURE vs what they DERIVE

- **Direct measurement**: Physical property detected by sensor
- **Derived quantity**: Hydrological variable calculated using models

Example exam question format:

- Name 3 satellite missions
- Describe measurement principles
- What hydrological quantity is derived?
- How could similar measurements be made on the ground?

Key insight

Satellites rarely measure hydrological variables directly. They detect physical properties that are converted to water information using retrieval algorithms.

Ground-based alternatives:

- Point measurements (wells, sensors)
- Hydrological modeling
- Never at global scale like satellites

Key Satellite Missions: Measurement Principles

GRACE (Gravity Recovery and Climate Experiment):

- **Measures**: Inter-satellite distance changes
- **Derives**: Total water storage variations
- **Includes**: Groundwater, soil moisture, surface water
- **Ground alternative**: Local wells + modeling (not global)

SMAP (Soil Moisture Active Passive):

- **Measures**: Microwave brightness temperature
- **Derives**: Soil moisture content
- **Physical basis**: Dielectric properties of land surface
- **Ground alternative**: TDR probes, capacitance sensors

SMOS (Soil Moisture and Ocean Salinity):

- **Measures**: Microwave brightness temperature
- **Derives**: Soil moisture + ocean salinity
- **Physical basis**: Dielectric properties
- **Ground alternative**: Point-scale sensors only

Key point for exam:

- Understand the **physical measurement**
- Know what **hydrological quantity** is derived
- Recognize **limitations** of ground-based alternatives

Orbital Characteristics: Impact on Hydrological Monitoring

Orbit inclination determines coverage:

- **Polar orbits**: Global coverage, sun-synchronous
- **Geostationary**: Continuous regional observation
- **Low inclination**: Limited latitudinal coverage

Altitude effects:

Trade-offs for hydrological applications:

Parameter	Low Orbit	High Orbit
Resolution	High	Low
Coverage	Limited	Global
Revisit time	Frequent	Less frequent
Mission life	Short	Long

Table 7

Exam focus: Understand how orbital design optimizes missions for different water cycle components

- **Lower altitude** (few hundred km):
    - High spatial resolution
    - Higher atmospheric drag
    - Shorter mission lifetime
- **Higher altitude**:
    - Broader coverage
    - Lower resolution
    - Longer mission duration

Satellite Fingerprints: Mission Optimisation

Key parameters for hydrological monitoring:

- **Swath width**: How much area covered per pass
- **Spatial footprint**: Size of individual measurement
- **Accuracy**: Precision of derived quantities
- **Temporal revisit**: How often same area observed

Mission optimisation examples:

- **Precipitation**: High temporal resolution needed
- **Soil moisture**: Moderate resolution, global coverage
- **Storage (GRACE)**: Low resolution, global coverage
- **Surface water**: High resolution, frequent revisits

Exam strategy

For each mission, be able to explain:

1. **What** it measures directly
2. **What** hydrological quantity it derives
3. **Why** this approach is used
4. **How** ground measurements differ

Critical understanding: Different missions optimized for different water cycle components based on their measurement capabilities and orbital characteristics.

Summary: Groundwater Fundamentals

Key takeaways from this overview:

- **Fundamental parameters** (porosity, K, T, S) control groundwater behaviour
- **Darcy's Law** provides the foundation for all flow analysis
- **Aquifer types** have distinct characteristics and management challenges
- **Unsaturated zone** adds complexity through variable properties
- **Flow theory** enables quantitative analysis and prediction
- **Real-world applications** require integration of multiple concepts and careful consideration of site-specific conditions
- **Satellite geodesy** provides global-scale water monitoring through indirect measurements and retrieval algorithms